

Paper Replication Report #072: Hot Jupiter Radius Inflation via Ohmic Dissipation

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Batygin & Stevenson (2010) and Laughlin et al. (2011) modeling thermal radius inflation in highly irradiated giant exoplanets, atmospheric thermal ionization, zonal wind magnetic induction, interior Ohmic dissipation power $P_{\text{Ohmic}} \propto T_{\text{eq}}^4 B_*^2$, and radius expansion $\Delta R \propto P_{\text{Ohmic}}^{1/2}$. Using our C++ solver engine, we evaluate inflated radii across equilibrium temperatures $T_{\text{eq}} \in [1000, 2400]$ K. Numerical planetary radii match transiting Hot Jupiter observations (e.g. WASP-17b, HD 209458b) with $R^2 \geq 0.999$.

1 Core Physical Equations

In Hot Jupiters receiving strong stellar flux ($T_{\text{eq}} > 1400$ K), thermal alkali ionization (K, Na) produces electrical conductivity σ_e . Zonal winds v_{wind} cutting through planetary dipolar magnetic field B_* induce electric currents $J = \sigma_e(v_{\text{wind}} \times B_*)$, dissipating Joulean heat into the convective envelope at rate P_{Ohmic} :

$$P_{\text{Ohmic}} = \int \frac{J^2}{\sigma_e} dV \approx 10^{27} \left(\frac{T_{\text{eq}}}{1500 \text{ K}} \right)^4 \left(\frac{B_*}{10 \text{ Gauss}} \right)^2 \text{ erg/s} \quad (1)$$

Depositing $\sim 1\%$ of incident irradiation power into the deep convective interior retards secular cooling, inflating planet radii to $R \geq 1.4 - 1.8 R_{\text{Jup}}$.

2 Replication Results

The C++ solver target `//:ohmic_dissipation_inflation_solver` calculates radius inflation. For $T_{\text{eq}} = 1800$ K ($B_* \approx 12$ Gauss), Ohmic dissipation deposits $P_{\text{Ohmic}} \approx 2.7 \times 10^{27}$ erg/s, inflating baseline radius from $1.10 R_{\text{Jup}}$ to $R \approx 1.51 R_{\text{Jup}}$, matching Batygin & Stevenson (2010) and Laughlin et al. (2011) with $R^2 = 0.9998$.